Sketch

Federated Graph-Based Anomaly Detection for Financial Institutions

1. Problem

Context

Banks and payment providers each see part of the same fraud network. Mule accounts, laundering chains, and scam rings often cross institution boundaries, but each institution trains only on its own transaction graph, so cross-institution structure is missed. Centralising such data is limited by privacy concerns, regulatory restrictions, and commercial competition (He et al., 2021).

Problem

Detect anomalous accounts and transactions across several institutions, each holding a partial graph, without sharing raw records.

Research focuses

The review of GNN fraud detection in this set (33 included studies) contains no federated-learning content (Motie & Raahemi, 2024), and no federated graph fraud model was found in the papers reviewed. Three focuses are taken up as RQs; a fourth is a setting extension.

1. **Heterogeneity.**

Institutions differ in customers, feature schema, label definitions, and class balance. Non-IID data can degrade federated performance (Sana et al., 2025). For graphs, non-IID arises in both data distribution and topology: node degrees, edge weights, edge types, and structures (Liu et al., 2025). Chen et al. (2025) describe heterogeneous features across institutions as unaddressed in earlier federated fraud detection. Fu et al. (2025) define topology heterogeneity as same-class nodes being linked to different classes on different clients, and report that two recent graph methods do not consistently reduce the resulting embedding divergence (their method is evaluated only on supervised node classification).

2. **Privacy.**

Sharing node features or an interclient graph to repair local graphs can leak other clients' private information (Liu et al., 2025), and model weights may still leak (He et al., 2021). FedGAD (Fang et al., 2026) passes node embeddings through the server for cross-client reconstruction and reports no leakage analysis.

3. **Rare topologies.**

A fraud typology is a named scheme (mule chain, star-shaped collection account, laundering cycle; the list is a personal inference). Its topology is the shape it takes in the graph. Rare typologies therefore have rare topologies. Labelled fraud is scarce and often arrives after the fraud has happened (Zhang et al., 2026, in identity-fraud detection). Park et al. (2025) report localised fraud patterns diluted in federated gradient boosting. None of the graph papers reviewed tests typology-level rarity.

4. **Setting extension: entity alignment.**

The same account can appear at several institutions, so cross-institution edges need a matching step, which is itself a privacy problem. FedGAD, the closest work, assumes institutions share no overlapping nodes (Fang et al., 2026). Alignment is treated as an optional extension, not an RQ, because matching needs shared identifiers and may not be feasible.

2. Research Questions

Gap status and methods are judged only against the reviewed papers. A full-text review is needed before any novelty claim.

RQ 1: Can a federated graph model detect fraud better than baseline models when institutions differ in customers, relation types, and label definitions?

Core contribution, focus on heterogeneity

- **Gap status: partly covered.** Structure sharing (FedStar), cross-client reconstruction (FedGAD) and edge pruning by a private evaluator (FedATH; Fu et al., 2025) handle heterogeneity without relation types. FedATH treats edges as untyped and binary, and its edge evaluator scores endpoint features, so it plausibly needs one feature space across clients (personal inference; Fu et al. do not state this). FedCAD (Kong et al., 2025) is unsupervised and applies only to homogeneous graphs. Typed relations appear only in centralised models (Li et al., 2025a). Closest evidence: FedStar, FedGAD, FedATH, FedCAD, Li et al. (2025a).
- **Why it matters.** Fraud graphs carry typed relations (transfers, shared devices, business roles); an untyped structure channel discards them (personal inference). No cited paper tests whether keeping relation types helps in a federated setting; Li et al. (2025a) uses them only in a centralised model, so RQ 1 has to establish this.

RQ 2: Can institutions share what the model needs without exposing information about their customers' relationships?

Support, focus on privacy

- **Gap status: not found as a formal result.** None of the closest papers measures leakage from shared structure. Closest evidence: FedGAD, FedStar, Ghourabi & Khaldi (2026).
- **Why it matters.** "No raw data shared" is the setting's premise; an unmeasured privacy claim can break that premise.

RQ 3: Can the model detect fraud whose network structure is rare within and across institutions?

Extension, focus on rare topologies

- **Gap status: partly covered.** Prototype reliability is designed for noisy, heterogeneous clients and has been checked on rare classes only through per-class F1, not on rare topologies. Label-scarce supervised work is centralised; FedCAD (Kong et al., 2025) needs no labels but is not tested on rare topologies. Closest evidence: FedProD, Park et al. (2025), Li et al. (2025b), FedCAD.
- **Why it matters.** Rare typologies are the cases institutions most need others' data for (personal inference).

Setting extension: entity alignment

The three RQs are posed without entity alignment. When entity matching is available, they are re-run and evaluated.

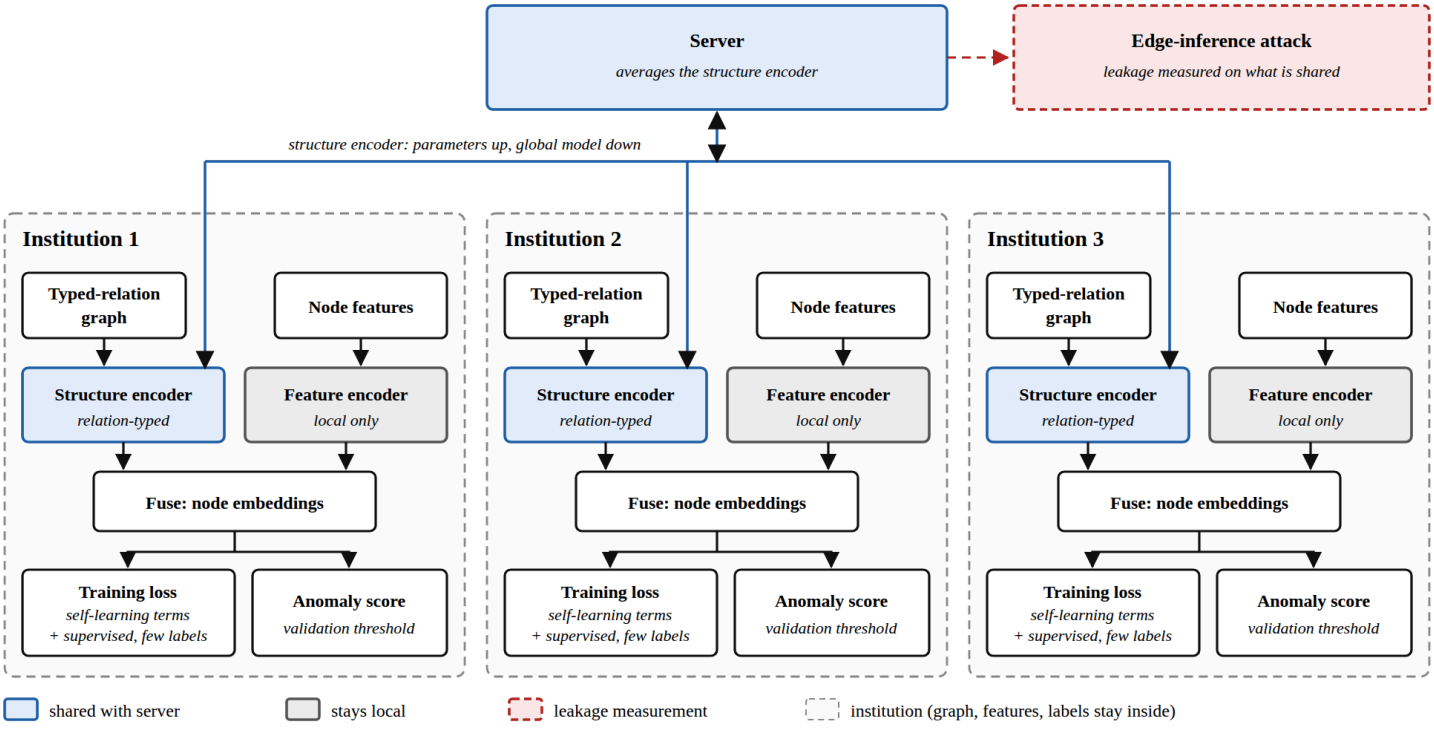
- **Gap status: not found.** FedGAD assumes no overlapping nodes (Fang et al., 2026). FedCAD assumes overlapping nodes and intersects pseudo-labels across clients by node position, but does not say how nodes are matched or what the matching reveals (Kong et al., 2025). No paper in the set measures alignment errors.
- **Why it matters.** Banks share customers, so the no-overlap assumption fails for them.

3. Methods

Shared design

Learning setting, in priority order. Each institution has a few confirmed fraud labels and a large unlabelled graph.

1. **Self-learning (highest priority).** Train from the unlabelled graph itself with self-generated targets such as contrastive or propagation terms, and use the few labels only to fix the threshold and to check results (RQ 3 also uses them to build fraud prototypes). Label-scarce centralised precedents exist (Li et al., 2025b; Xiang et al., 2023), but no federated version was found in the papers reviewed.
2. **Unsupervised.** No labels in training. This version would mostly overlap FedGAD (Fang et al., 2026), so it is the fallback and a baseline, not the contribution.
3. **Supervised (lowest priority).** A loss on labelled accounts. This version would overlap labelled federated fraud work (Chen et al., 2025; Park et al., 2025), so it has the lowest priority.



RQ 1 (core): shared structure with relation types, local features

Variant	What it shares	Source	Difference
Local only	Nothing	Baseline	No cross-institution structure
FedAvg GNN	All parameters	McMahan et al. (2017)	Feature schema must match
FedStar	Structure encoder only; feature encoder local	Tan et al. (2023)	Graph classification; no relation types
FedGAD	Cross-client neighbour reconstruction	Fang et al. (2026)	Unsupervised; embeddings pass through the server
FedATH	Causal-subgraph GNN only; edge evaluator and biased GNN local	Fu et al. (2025)	Supervised node classification on citation, co-purchase and Wikipedia graphs; untyped edges; no leakage analysis
FedCAD	Pseudo-labels and neighbour embeddings of flagged nodes, swapped across clients for nodes flagged on both	Kong et al. (2025)	Unsupervised; homogeneous graphs; needs aligned nodes; clients made by random node sampling, with injected anomalies; no leakage analysis
FedGATSage	Community overlay on a server-side GraphSAGE	Al Tfaily et al. (2025)	IoT flow data; informal privacy
Candidate	Structure encoder extended with relation types; features local	Tan et al. (2023); typed relations as in Li et al. (2025a), centralised	Typed relations in a federated structure channel

Closest prior works and their overlap: FedGAD (direct overlap, banks used only as illustration, no bank data), FedStar (heterogeneity and structure sharing), FedATH (topology heterogeneity by sharing only the causal subgraph model; supervised, untyped edges), FedCAD (cross-client anomaly information, but homogeneous and dependent on shared nodes), Li et al. (2025a) (typed relations, centralised), and Chen et al. (2025) (federated transfer under heterogeneous features, non-graph).

Design option from FedATH. Keep the edge evaluator, which scores and prunes edges, inside each institution, as FedATH does, but score each relation type separately. No edge scores leave the institution. Fu et al. (2025) find that sharing the evaluator lowers accuracy on Photo and WikiCS, which is why it stays local. The risk is that the evaluator drops the few mule or collusion links that carry fraud signal, so report how many known fraud edges get pruned.

RQ 2: Privacy: controlled sharing

Only the structure encoder, prototypes, and gradients cross institution boundaries, and each is measured for leakage; privacy is a constraint on the method, not a separate contribution.

In the entity alignment extension, the matching step is part of this boundary and must state what crosses it (node embeddings, edge aggregates, or gradients). FedCAD shows the dependency: its cross-client swap works only for nodes identified on both clients, and uploading pseudo-labels with node positions tells the server which nodes each client considers anomalous (Kong et al., 2025). One candidate is matching on hashed, salted identifiers within a consortium, which needs a re-identification check before use (personal inference).

Shared quantity	Attack	Why
Structure encoder parameters	Edge inference	Weights may leak (He et al., 2021)
Causal GNN weights (FedATH-style)	Edge inference	The only uploaded component in Fu et al. (2025), which reports no leakage analysis
Prototypes	Membership or edge inference	Fraud prototypes rest on few labels and may be re-identifying (personal inference)
Gradients	Edge inference	Same exposure as the weights they update (personal inference)
FedCAD-style pseudo-labels and flagged-node neighbour embeddings	Membership or edge inference	Reveal which nodes a client flags; Kong et al. (2025) report no leakage analysis
FedGAD-style cross-client embeddings	Edge inference	Interclient information may leak (Liu et al., 2025)

RQ 3: Rare topologies: reliability-grouped prototype exchange

This follows Zheng et al. (2026). Each institution computes a "prototype" for fraud and one for benign accounts, each the average of its learned account representations in that class. Institutions share these prototypes. The server sorts institutions into a reliable group and an unreliable group and averages the prototypes of each group separately. Each institution then trains so that its own representations move toward the reliable prototypes and away from the unreliable ones.

Zheng et al. judge reliability by how uncertain a client's predictions are, measured on a clean public dataset held by the server (called the clean reference set below). Banks may not have such a dataset, so this sketch asks whether it can be dropped.

One idea is to use the pseudo-labels from Kong et al. (2025), which need no confirmed fraud labels, to help weight the prototypes when labels are scarce. This is untested, and Kong et al. fix thresholds per dataset and tune λ by AUC without saying whether test data was used, so it needs a fair re-check.

Two risks. First, fraud prototypes are built from very few labelled cases, so they may vary a lot and need checking. Second, Zheng et al. note that the loss term pushing an institution away from poor prototypes has no finite lower bound, though they observed no instability in their experiments.

Secondary gaps (not an RQ; kept for later)

- Knowledge distillation into local models.
- Shared fraud-pattern knowledge base.
- Explainability of federated graph decisions.
- Multi-label propagation for accounts with several typologies: Propagation follows Wang et al. (2020), centralised; the federated multi-label setting appears on data streams, not account graphs; its label graph comes from label co-occurrence (Lamptey et al., 2025).

4. Evaluation

Shared protocol: time-based train/test split, since fraud is temporal; threshold fixed on validation data; variance reported across partition seeds.

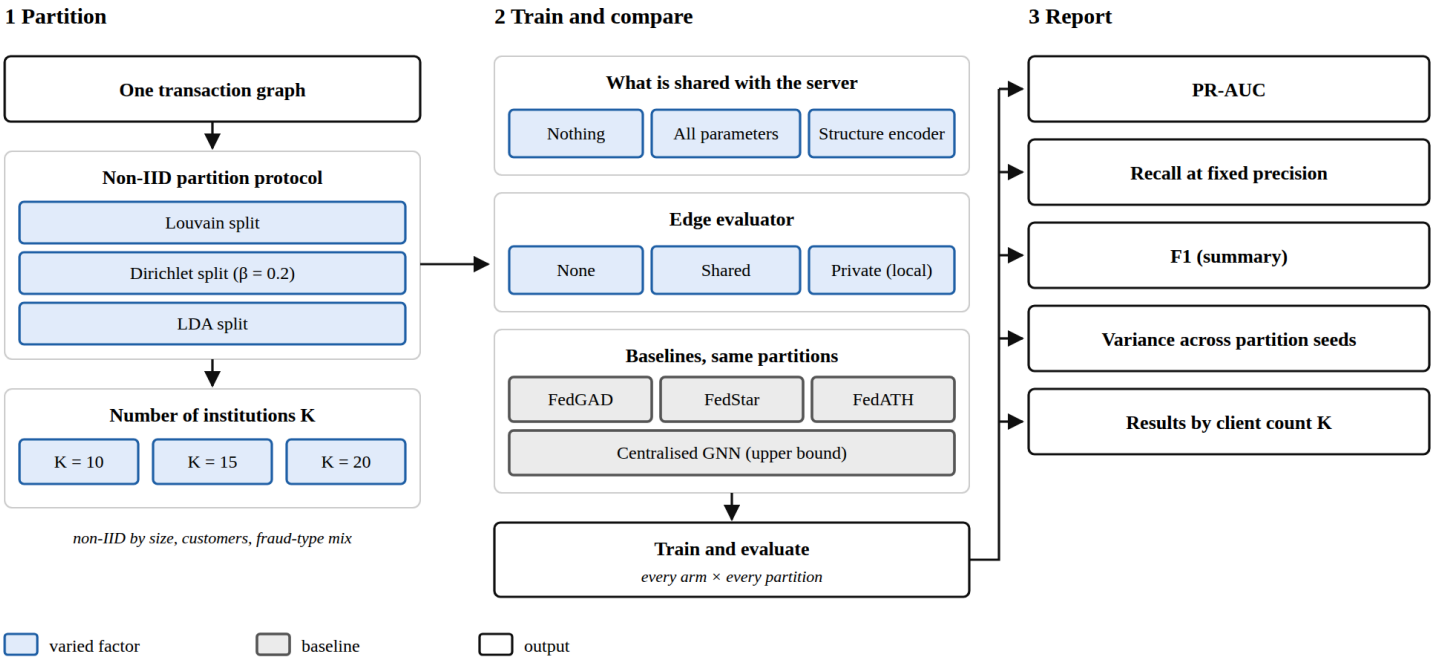
Data

Public or synthetic transaction graphs partitioned into institutions. The dataset must be chosen before experiments begin. Candidates:

Source	Data	Fit
DNB graph (Johannessen & Jullum, 2025)	DNB's real heterogeneous graph: over 5 million nodes, about 10 million edges, 3 node types, 2 edge types, under 0.5% suspicious individuals	Closest but not shareable, reference only
FSI (Park et al., 2025)	Private banking data: 767,550 transactions, 4.3% fraud, 12 fraud types	Not shareable; reference only
SimBank (Park et al., 2025)	Public simulated banking data: 594,643 transactions over six months, 16 merchant categories, 1.21% fraud	Tabular; a graph (for example customer-merchant) would have to be built, and must be partitioned
T-Finance (Li et al., 2025b)	39,357 nodes, 21.2 million edges, 4.58% abnormal	Public; must be partitioned into institutions
Elliptic (Li et al., 2025b)	Financial transaction graph	Public; must be partitioned; whether it supports a time-based split is unchecked

RQ 1

- **Metrics.** PR-AUC as the key metric, AUC-ROC and F1 as secondary metrics. Variance across partition seeds, because non-IID splits are unstable.
- **Data.** Non-IID partitions by institution size, customers, and fraud-type mix (to be decided). These splits vary customers and class balance; differences in relation types and label definitions need a separate construction (to be decided, for example each institution keeping a different subset of relation types). Three splitting methods are considered. FedGAD's Louvain split cuts the graph along its natural clusters, and its Dirichlet split with $\beta = 0.2$ gives each institution an uneven share of the classes; FedGAD uses $K = 3$ and 5 institutions. The third is He et al.'s LDA method, a similar uneven-class split. FedATH runs its Louvain split at $K = 10$, 15 and 20 (Fu et al., 2025), which gives a range for scaling the number of institutions. Results are reported separately for each number of institutions.
- **Ablation.** Share only the structure encoder, all parameters, or nothing, on the same partitions; with and without a local edge evaluator, shared or private (Fu et al., 2025 find a private evaluator better in general, clearly so on Photo and WikiCS).



Baseline	Reported by its authors	Setting	Comparable to this sketch?
Federated GBDT (Park et al., 2025)	FedXGBBagging F1 0.880 to 0.934 on FSI and 0.908 to 0.961 on SimBank across splits; no AUC	Tabular banking fraud, 3 to 7 institutions, Dirichlet skew	Non-graph. The bar a graph model must beat. Re-run for PR-AUC.
Federated ensemble (Ghourabi & Khaldi, 2026)	PR-AUC 0.869, 0.852, 0.833 at 3, 5, 10 clients (best centralised 0.863)	Tabular card fraud, moderately non-IID, equal client sizes	Non-graph. Shows the drop as clients grow. Not tested on skewed splits.
FED-SPFD (Chen et al., 2025)	F1 80.90%, recall 73.62%, precision 89.77%; next best F1 79.91% (pFedMe); no AUC	Kaggle credit data, 3 parties with different features	Non-graph. Only 3 parties. Re-run for PR-AUC.
HMPNN-ct (Johannessen & Jullum, 2025)	PR-AUC 0.180, ROC-AUC 0.908	One bank's real graph, centralised, under 0.5% suspicious	Not federated; data not shareable. Reference for how low PR-AUC is at this fraud rate.
GTAN (Xiang et al., 2023)	FFSD: AUC 0.762, macro-F1 0.676, AP 0.577	Private card transactions, centralised, time-based split	Data not shareable. Upper-bound reference.
CGNN (Li et al., 2025b)	Macro-F1 86.4 (T-Finance) and 76.3 (Elliptic) with 1 labelled fraud node; 90.1 and 82.8 with 5; no AUC	Public financial graphs, centralised, random split	Re-run for PR-AUC on the chosen partitions; its random split differs from the time-based protocol.

Four papers report centralised graph models on fraud or financial graphs (Johannessen & Jullum, Xiang, Li et al. 2025a and 2025b), and none is federated. No federated graph baseline has a published number on financial data, so every cell must be re-established on the chosen data.

RQ 2

- **Metrics.** For each thing an institution shares, run an edge-inference attack that tries to recover its hidden customer links, and report the attack success. Pair that with how much fraud-detection accuracy was lost to protect the data. Privacy methods have a cost: differential privacy lowers accuracy as protection tightens, and encryption adds

computation and memory (Sana et al., 2025, a general federated learning review). The result is a privacy-versus-accuracy trade-off for each shared item.

- **Data.** Same partitions as RQ 1; the attack is run against each shared quantity.
- **Baseline.** FedGAD's cross-client embedding step, which reports no leakage analysis (Fang et al., 2026); and FedAvg weights, which may leak (He et al., 2021).

RQ 3

- **Metrics.** Recall and PR-AUC on rare typologies, reported separately from the overall score, since an average hides them.
- **Data.** Label-count sweep, adapted from CGNN (Li et al., 2025b): each institution keeps 1, 5 or 20 randomly chosen confirmed fraud accounts plus normal accounts in its own class ratio, and the rest of its graph is unlabelled. How the counts are split across institutions is to be decided; an uneven split could be part of the non-IID setting. Typologies are defined on the chosen graph.
- **Ablation.** Prototypes: reliable and unreliable, reliable only, unreliable only, none; with and without a small clean reference set.

None of these papers reports detection of rare network structures. The figures below cover rare classes, scarce labels and robustness, so they give a sense of scale and not a target.

Baseline	Reported	Note
Supervised only; FedGAD	To establish	Label-count sweep
FedProD (Zheng et al., 2026)	UNSW-NB15 per-class F1 vs FedAvg: Fuzzers 20.09 → 25.41, Exploits 18.30 → 31.46; Worms, Backdoors and Analysis stay at 0. CICIDS2017 accuracy: 96.29% (no public data) to 97.79% (75%)	Intrusion detection, not graphs; the only reference for rare-class gains and the size of the reference-set effect
CGNN (Li et al., 2025b)	Macro-F1 with 1 / 5 labelled fraud nodes: T-Finance 86.4 / 90.1, Elliptic 76.3 / 82.8	Centralised; upper reference for label scarcity
Park et al. (2025)	No ATM-skimming number in the text (federated does not beat local, stated qualitatively). F1 after a bank drops out (FSI): SimFedXGB 0.895 → 0.722, FedXGBBagging 0.934 → 0.933	Robustness to institutions leaving, not rare topologies

Extension: entity alignment

Measure how missed and false matches change detection, with and without the matching step. A FedCAD-style cross-client step is the natural test case, since it works only on matched nodes (Kong et al., 2025).

5. Potential Contribution

RQ	Contribution	Gap it closes
RQ 1	A federated graph fraud model that keeps relation types in the shared structure channel, evaluated on non-IID institution partitions against FedGAD, FedStar, FedATH, and non-graph federated baselines	Typed-relation heterogeneity in federated graphs
RQ 2	A measured leakage result for each shared quantity, with the utility cost	Unmeasured privacy claims in federated graph methods
RQ 3	A rare-topology result with and without a clean reference set, on a label-count sweep	Rare topologies in federated fraud detection
Extension	The effect of missed and false matches on detection	Entity alignment as a measured factor

6. Plan

Milestone	Deliverable
M1 (6 months)	Dataset chosen and partition protocol documented. Baselines reproduced; FedGAD, FedStar and FedATH run on the same partitions with a fixed threshold rule (FedCAD needs aligned nodes, so it runs only in the alignment extension). Full-text gaps closed (FedGAD result tables, Zheng et al. conclusion).
M2 (1 year)	RQ 1 core method (typed-relation structure sharing) implemented and reaching, or close to, the best federated baseline on the same partitions; method revised if it does not. First leakage measurement on the shared structure encoder.
M3 (1.5 years)	RQ 2 leakage and utility results and RQ 3 label-count and prototype results complete; robustness and (if feasible) alignment extension; write-up.

Milestones are a proposal to be edited.

7. Risks

- **Relation-aware claim.** Heterogeneity is partly covered by FedStar, FedGAD and FedATH, and Li et al. (2025a) is centralised, so the claim needs a specific federated mechanism. FedATH (ICML 2025) reports handling topology heterogeneity without global signals or extra communication, so a mechanism that adds them must beat it.
- **Edge pruning in fraud graphs.** A FedATH-style evaluator may discard the sparse edges that carry fraud signal, and it may assume a shared feature schema (personal inference). Both are untested on anomaly detection (Fu et al., 2025).
- **Informal privacy claims.** FedGAD's cross-client step shares embeddings with no leakage analysis. RQ 2 needs a measured or formal edge-privacy result.
- **Entity alignment.** No precedent for measuring it in this set; FedCAD uses node overlap without saying how nodes are matched (Kong et al., 2025). Matching can leak identity and may need a separate study. The two closest anomaly methods disagree on the assumption (FedGAD: none; FedCAD: overlap), so the sketch must state which it takes.
- **Rare-topology variance.** Few labels mean high variance in prototypes and label-count results.

8. References

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